Candidate Seat no………….

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech (EC) Examination

May 2018

EC311.01/EC311 VLSI Technology and Design

Date: 08/05/2018(Tuesday) **Time: 10:00 am to 01:00 pm** **Maximum Marks: 70**

Instructions:

1. The question paper comprise of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 **Answer the questions below.** **[07]**

(a) Latch-up condition is observed in _______. **[01]**

I. Resistive Load Inverter II. Depletion Load Inverter

III. CMOS Inverter IV. None of these

(b) Regularity is a VLSI Design strategy in which a module is sub-divided to max number of similar sub-modules at each level. **True /False** **[01]**

(c) Which of the following is true for critical voltage (V_{OH} ,V_{OL}, V_{IH}, V_{IL}) of an inverter design ? **[01]**

I. Noise Margin depends on V_{OH} ,V_{OL}, V_{IH}, $V_{IL.}$

II. Noise Margin depends only on V_{OH} , $V_{OL.}$

III. Noise Margin depends only on V_{IH} ,$V_{IL.}$

IV. Noise Margin is independent of the critical voltages.

(d) In NMOS transistor, the substrate is of ______ type and source & drain are heavily doped with ______ impurity. **[01]**

I. N , Pentavalent II. P , Trivalent

III. N , Trivalent IV. P , Pentavalent

(e) The design flow of VLSI system is__ **[01]**

1. Architecture design 2. Market requirement 3.Logic design 4.HDL coding

I.2-1-3-4 II.4-1-3-2 III.3-2-1-4 IV.1-2-3-4

(f) Which of the following is a correct combination of threshold voltages of Depletion type NMOS transistor and Enhancement type PMOS transistor respectively? **[01]**

I. 0.7V, 0.7V II. 0.7V, -0.7V III. -0.7V, -0.7V IV. -0.7V, 0.7V

(g) Which of the following VLSI design strategy ensures maximum local connections such that keeping critical paths within module boundaries? **[01]**

I. Hierarchy II. Regularity

III. Modularity IV. Locality

Q - 2 Answer the following questions. **[14]**

(a) Differentiate between Enhancement type and Depletion type MOSFET and also draw symbol of both type transistor. **[02]**

(b) Realize the CMOS logic circuit for $\mathbf{Z} = \overline{A.(B+C)+D.E}$. **[06]**

Find the equivalent CMOS inverter circuit for simultaneous switching of all inputs, assuming that $(W/L)_p = 10$ for all pMOS transistors and $(W/L)_n = 15$ for all nMOS transistors.

(c) State the significance of Channel Length Modulation. Explain channel length modulation in detail. **[06]**

OR

(c) Explain MOS capacitor under external bias with energy band diagrams. **[06]**

Q - 3 Answer any two questions. **[14]**

(a) Explain Fabrication process of NMOS transistor with necessary diagram. **[07]**

(b) Compare effect of Semi-custom and Full custom VLSI design style on chip performance. Also compare effect of Manufacturing Technology window on chip performance. **[07]**

(c) Design 3 staged depletion-load nMOS dynamic shift register. Explain its operation. **[07]**

SECTION – II

Q - 4 Answer the questions below. **[07]**

(a) Which of the following statement(s) is/are true in context to CMOS Inverter? **[01]**

I. Steady State Power Consumption is negligible

II. Full Output Voltage swing

I. Statement I only II. Statement II only

III. Both Statement I & Statement II IV. Neither Statement I nor Statement II

(b) What is the need of stick diagram? **[02]**

(c) Implement NOR based SR Latch using CMOS Logic. **[02]**

(d) Implement XNOR gate using CMOS Transmission Gates. **[02]**

Q - 5 Answer the following questions. **[14]**

(a) CMOS ring oscillator where each stage is having propagation delay of 20 ns. Calculate oscillation frequency of a Three stages. **[02]**

(b) Implement D Latch using CMOS transmission gates and CMOS inverter. **[06]**
Explain race around condition (toggle switch problem) of J-K Latch. Also indicate methods to overcome this problem.

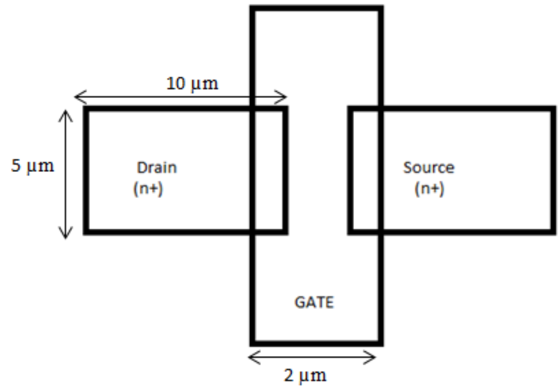
(c) Consider the following MOSFET process. **[06]**

Substrate doping $N_A = 2*10^{15}$ cm^{-3}, $\varphi_{GC} = -0.85$ V , $Q_{ox} = q* 2* 10^{11}$ C/cm^2 ,gate oxide thickness t_{ox}= 200 Å, Calculate the threshold voltage V_{T0} for V_{SB} = 0 V.

Use $\epsilon_{ox} = 3.97\epsilon_0$, $\epsilon_{si} = 11.7\epsilon_0$, where $\epsilon_0 = 8.854\times10^{-14}$ F/cm.

OR

(c) Consider the enhancement NMOS Transistor and parameter are given as follows: **[06]**
Substrate doping: $N_A = 2 *10^{16}$ cm^{-3} , Source/Drain doping $N_D = 10^{19}$ cm^{-3}, Sidewell (P^+) doping $N_{A(sw)} = 4 *10^{16}$ cm^{-3}, gate oxide thickness t_{ox}= 45nm, Junction depth x_j= 1.0 µm. Drain voltage is varies from 0.5 V to 5 V.
Calculate the average drain-substrate junction capacitance C_{db}.



Q – 6 Answer the following questions. [14]

(a) Draw the circuit and Voltage transfer characteristics (VTC) of CMOS inverter. **[02]**

(b) Consider CMOS Inverter circuit with V_{DD}= 3.3 V. The I-V characteristics of the nMOS transistor are specified as follows: V_{GS}= 3.3 V, I_{sat}= 2 mA for $V_{DS} \geq 2.5$ V. Assume input signal applied to gate is a step pulse switching from 0 V to 3.3 V. Consider $V_{T,n}$=0.8 V, C_{load}= 300 pF. Calculate τ_{phl} using static equation method. **[06]**

(c) What is the need for voltage bootstrapping? Explain dynamic voltage bootstrapping circuit with necessary mathematical analysis. **[06]**

OR

(c) Explain the principle of dynamic CMOS logic (Precharge - Evaluate logic).Discuss its advantages and disadvantage. **[06]**